Solving Linear Systems

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1 Introduction

Many problems in numerical analysis, engineering, and scientific computing can be expressed as a system of linear equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n. \end{aligned}$$

This system can be written compactly in matrix form as

$$A\mathbf{x} = \mathbf{b},$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Here,

- A is the **coefficient matrix**,
- $\mathbf{x}$ is the **unknown vector**,
- $\mathbf{b}$ is the **right-hand-side vector**.

The objective is to determine $\mathbf{x}$ such that $A\mathbf{x} = \mathbf{b}$.

1.1 Direct and Iterative Methods

Direct methods obtain the solution through a finite sequence of arithmetic operations. In exact arithmetic, they produce the exact solution after a finite number of steps. Examples include Gaussian Elimination and LU Decomposition.

Iterative methods begin with an initial approximation $\mathbf{x}^{(0)}$ and repeatedly improve it

$$\mathbf{x}^{(0)} \to \mathbf{x}^{(1)} \to \mathbf{x}^{(2)} \to \cdots$$

until the approximation satisfies a chosen convergence criterion. Examples include Jacobi Method, Gauss-Seidel Method and Conjugate Gradient Method.

2 Gaussian Elimination

Gaussian Elimination is a **direct method** for solving a system of linear equations $A\mathbf{x} = \mathbf{b}$. The main idea is to transform the original system into an **upper triangular system**, which can then be solved using **back substitution**.

2.1 Forward Elimination

Consider the system

$$\begin{aligned} 2x + y - z &= 8, \\ -3x - y + 2z &= -11, \\ -2x + y + 2z &= -3. \end{aligned}$$

Its augmented matrix is

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ -3 & -1 & 2 & -11 \\ -2 & 1 & 2 & -3 \end{array}\right].$$

The goal of forward elimination is to eliminate the elements below the diagonal, transforming the matrix into the form

$$\left[\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right].$$

First Pivot : The first pivot is $a_{11} = 2$.
To eliminate $a_{21} = -3$, calculate the multiplier

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{-3}{2}.$$

Then perform the row operation

$$R_2 \leftarrow R_2 - m_{21} R_1.$$

Similarly, to eliminate $a_{31} = -2$,

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{-2}{2} = -1,$$

and

$$R_3 \leftarrow R_3 - m_{31} R_1.$$

This produces

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 2 & 1 & 5 \end{array}\right].$$

The second pivot is $a_{22} = \frac{1}{2}$.
To eliminate $a_{32} = 2$, calculate

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{2}{1/2} = 4.$$

Then perform

$$R_3 \leftarrow R_3 - 4R_2.$$

We obtain

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 8 \\ 0 & \frac{1}{2} & \frac{1}{2} & 1 \\ 0 & 0 & -1 & 1 \end{array}\right].$$

The coefficient matrix is now in **upper triangular form**.

2.2 Back Substitution

The resulting system is

$$\begin{aligned} 2x + y - z &= 8, \\ \frac{1}{2}y + \frac{1}{2}z &= 1, \\ -z &= 1. \end{aligned}$$

Starting from the last equation,

$$z = -1.$$

Substitute $z = -1$ into the second equation:

$$\frac{1}{2}y + \frac{1}{2}(-1) = 1,$$

which gives

$$y = 3.$$

Finally, substitute $y = 3$ and $z = -1$ into the first equation:

$$2x + 3 - (-1) = 8,$$

so

$$x = 2.$$

Therefore,

$$\boxed{\mathbf{x} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}}.$$

2.3 Partial Pivoting

During forward elimination, the multiplier is calculated as

$$m_{ik} = \frac{a_{ik}}{a_{kk}},$$

where a_{kk} is the current **pivot**. If the pivot $a_{kk} = 0$, the multiplier cannot be calculated. Furthermore, if a_{kk} is very close to zero, division by a small number may amplify round-off errors and lead to numerical instability.

To avoid this problem, Gaussian Elimination is commonly implemented with **partial pivoting**. For each pivot column k, partial pivoting selects the element with the largest absolute value among the entries at or below the current pivot position:

$$|a_{pk}| = \max_{i=k,\ldots,n} |a_{ik}|.$$

If $p \neq k$, rows k and p are exchanged:

$$R_k \leftrightarrow R_p.$$

For example, consider

$$\left[\begin{array}{cc|c} 0.001 & 1 & 1 \\ 5 & 2 & 7 \\ -2 & 3 & 1 \end{array}\right].$$

In the first column,

$$|5| > |-2| > |0.001|,$$

so the second row is selected as the pivot row. After exchanging R_1 and R_2,

$$\left[\begin{array}{cc|c} 5 & 2 & 7 \\ 0.001 & 1 & 1 \\ -2 & 3 & 1 \end{array}\right].$$

The elimination can then proceed using 5 as the first pivot. Partial pivoting improves the numerical stability of Gaussian Elimination and avoids division by zero when a suitable nonzero pivot exists.

2.4 General Algorithm

For a general system

$$A\mathbf{x} = \mathbf{b},$$

Gaussian Elimination with partial pivoting consists of **forward elimination** followed by **back substitution**.

Forward Elimination

For each pivot column $k = 1, 2, \ldots, n-1$, perform the following steps.

1. **Select the pivot row**

   Find the row p containing the largest absolute value in the current pivot column:

   $$p = \underset{i=k,\ldots,n}{\operatorname{argmax}} |a_{ik}|.$$

2. **Check the pivot**

   If $|a_{pk}| < \varepsilon$, the matrix is singular or nearly singular, and the elimination cannot safely continue.

3. **Swap rows**

   If $p \neq k$, exchange rows k and p: $R_k \leftrightarrow R_p$. The corresponding elements of the right-hand-side vector $\mathbf{b}$ must also be exchanged.

4. **Eliminate the entries below the pivot**

   For $i = k+1, \ldots, n$, calculate the multiplier $m_{ik} = \frac{a_{ik}}{a_{kk}}$.

   Then update the elements in row i:

   $$a_{ij} \leftarrow a_{ij} - m_{ik}a_{kj}, \qquad j = k, k+1, \ldots, n,$$

   and update the corresponding element of the right-hand-side vector:

   $$b_i \leftarrow b_i - m_{ik}b_k.$$

After forward elimination, the coefficient matrix is transformed into an upper triangular matrix.

Before performing back substitution, the final pivot must also be checked. If $|a_{nn}| < \varepsilon$, the matrix is singular or nearly singular.

Back Substitution

Starting from the last equation,

$$x_n = \frac{b_n}{a_{nn}}.$$

The remaining unknowns are calculated from bottom to top:

$$x_i = \frac{b_i - \sum_{j=i+1}^{n} a_{ij} x_j}{a_{ii}}, \qquad i = n-1, n-2, \ldots, 1.$$